PCB Inspection Report

Timestamp	2026-04-04 16:37:41
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

1) Root Cause Analysis
The consistent detection of "Missing_hole" defects across seven distinct locations (confidence ranging from 0.7954 to 0.3038) strongly indicates a systemic issue within the drilling or drill data preparation processes. The primary contributor is likely an error in the CNC drill program, leading to specific holes being omitted or incorrectly defined. Alternatively, severe drill bit wear or breakage could have caused these omissions, particularly if the drill sequence grouped these locations. Material registration errors during drilling could also displace the intended hole locations, making them "missing" from their design intent. Post-drilling inspection system (AOI) calibration should also be verified, although the high confidence values for the top defects suggest they are genuinely absent rather than merely obscured or misdetected. The presence of lower confidence defects (e.g., 0.3384, 0.3038) might suggest partially formed holes, debris accumulation, or issues where the defect is less pronounced but still indicative of a drilling anomaly.
2) Operational Risks
The presence of multiple "Missing_hole" defects presents severe and immediate operational risks. If these holes are intended as plated through-holes (PTHs) for component leads or vias for signal routing, their absence will result in open circuits, rendering the PCB functionally inoperable. For non-plated through-holes (NPTHs) intended for mounting components, connectors, or heat sinks, the board will be unassemblable, or components will be insecurely mounted, leading to mechanical failure during operation due to vibration or stress. The high risk level and 70.06% failure probability are directly attributable to these fundamental structural and electrical omissions, preventing proper assembly and function of the end product.
3) Failure Prediction
Failure is predicted to be immediate and pervasive across the affected board. Given the 70.06% failure probability, the board is highly likely to fail initial electrical test (inability to complete circuits) or physical assembly (inability to mount components). If the missing holes are critical component

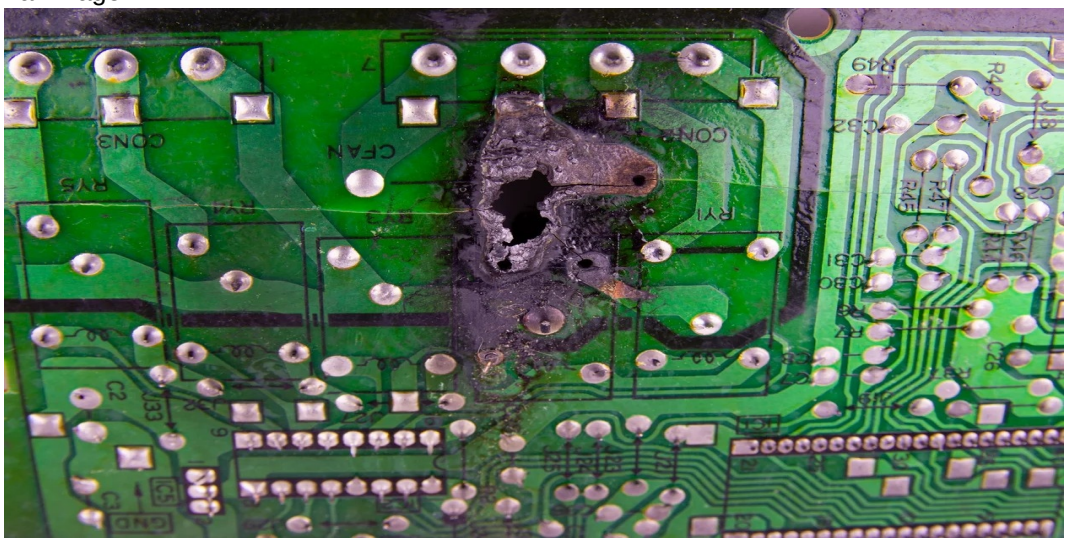
pin locations (e.g., BGA pads requiring vias), the component cannot be reliably attached. Should any missing holes correspond to power or ground vias, a complete functional failure is guaranteed. Even for less critical test points or non-structural holes, their absence indicates a critical deviation from design, precluding reliable operation or future diagnostics.
4) Repair Explanation
Due to the nature of "Missing_hole" defects, particularly when multiple and with high confidence, conventional repair is generally not feasible or recommended for production PCBs. Manually drilling holes into an already fabricated PCB poses significant reliability risks, including damage to internal layers, creation of drill burrs, contamination, and inability to plate the newly drilled holes (if PTHs were intended). This compromises electrical integrity, introduces potential shorts, and invalidates any subsequent reliability qualification. Therefore, for this defective PCB with systemic missing holes, the appropriate action is typically to scrap the board. Reworking would introduce unacceptable reliability compromises and is not a viable option for high-reliability applications.
5) Engineering Insights
The recurrence of "Missing_hole" defects across various locations necessitates an immediate hold on the current manufacturing lot and a thorough investigation. Key engineering insights involve:
a) Review of the Gerber drill file and comparison against the original design data to identify any discrepancies.
b) Examination of the drill program for errors or omissions.
c) Analysis of drill bit life, wear patterns, and breakage rates to assess tooling issues.
d) Verification of material registration on the drill machine to ensure proper alignment.
e) Evaluation of the AOI system's capability for detecting smaller or partially missing holes, especially given the lower confidence defects. This consistent defect type points to a fundamental process control issue within the drilling operation that requires immediate corrective action to prevent further defective output.

Repair Suggestions

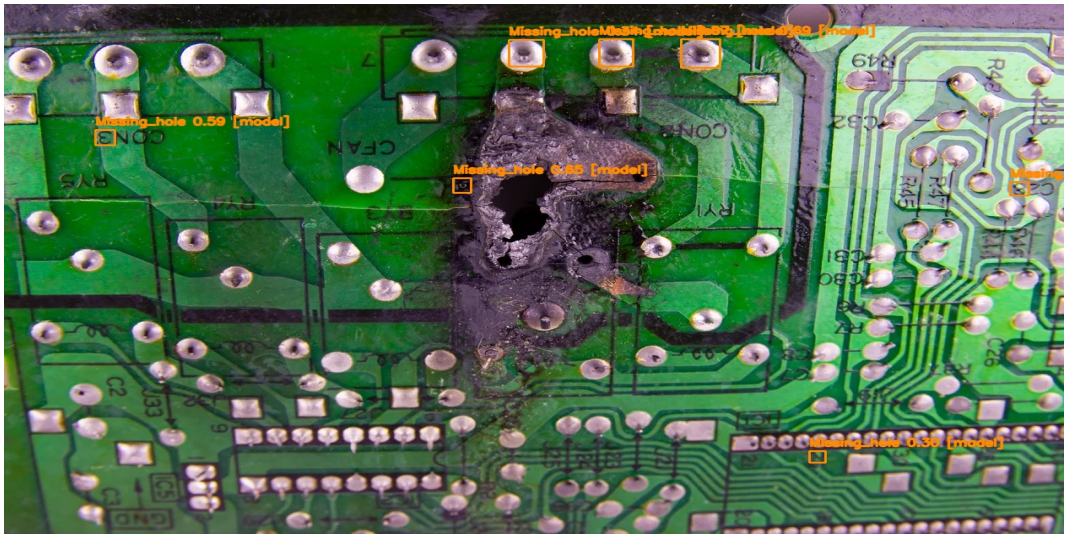
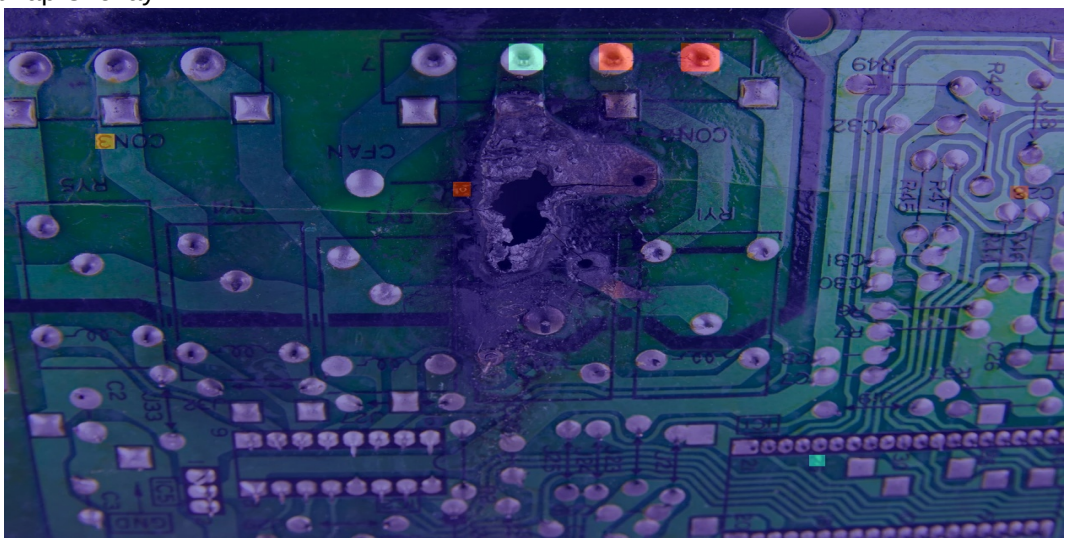
- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image



Detection Image



Heatmap Overlay